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User Research Log — Implementation Plan

Status: Sprint 1 shipped on dev 2026-05-16 (commit 11baa14). Sprints 2 + 3 scheduled. **Author:** Joe Pruskowski (with Claude) **Date:** 2026-05-16

Source entry: doc/Future_Ideas.md → “Research Log for users — in-platform training journal”.

Goal

Move the platform’s recommended “keep a research log” workflow **inside** the platform, so that:

1. **Users gain memory.** Every training session can carry structured notes the user can scan a month later.
2. **The platform gains training data.** Each note is paired with the hyperparameters, algorithm, and benchmark/ELO delta of the session that produced it. That triple — *free-form prose + structured config + outcome signal* — is exactly the shape a future reranker or fine-tune wants.
3. **The Help System gets smarter over time.** Published notes (opt-in) become a new corpus source. The retrieval pipeline mixes them in alongside the curated HelpDoc chunks, with attribution. Private notes are personal context only.

§1 Requirements

1.1 User-facing requirements

ID	Requirement
U1	Every Gym training session can carry at least one note (free-form text, ≤2 KB).
U2	Each note has an outcome label : success / plateau / regression / inconclusive.
U3	Each note has optional tags (free-form, comma-separated; normalized to lower-snake).
U4	Notes are private by default . A per-note “publish to community” toggle is the only way they become visible to others.
U5	Users can edit and delete their own notes. Editing a published note re-runs the PII scrub.
U6	The Gym → Sessions tab shows an inline “+ Add note ” affordance on every session row. Existing notes are listed under each row, newest first.
U7	A new Profile → Training journal tab shows every note the user has written, chronologically, with filters by algorithm / outcome / tag. Markdown rendering. Export to .md for backup.

ID	Requirement
U8	A user can also add ad-hoc entries (planning, retrospectives) not tied to a specific session. These live in the Profile journal but never in the Sessions tab.
U9	When the user asks the Guide a training question, the answer can cite their own past notes (“you wrote on 2026-04-12 ...”). Citation is gated by a Profile preference (shareNotesWithGuide, default ON).
U10	When a user opts in to publish a note, the answer can also cite <i>other users’ published notes</i> — with attribution shown as “Community note from @alice (2026-04-12)”.

1.2 Platform / system requirements

ID	Requirement
S1	Schema lives in the existing Prisma model (packages/db). Two new tables: TrainingSessionNote and ResearchLogEntry.
S2	The note layer does not duplicate hyperparameters or benchmark data — those already live on the existing TrainingSession row. Notes carry only the editorial content; reads JOIN.
S3	Published notes are mirrored into the Help corpus as a new HelpDoc.source = 'community-note'. HelpDoc.source is a new free-text column added in Sprint 2 (the plan originally assumed it already existed — drift found 2026-05-16). All pre-Sprint-2 rows backfill to 'guide'. Reindexing reuses the existing FTS + pgvector pipeline.
S4	The retrieval pipeline returns three categories of chunks per query: corpus (curated HelpDocs), private-notes (only the asking user’s), community-notes (published by others). The mixer assigns each category a budget so curated content always wins ties.
S5	A privacy scrubber runs on every publish action: regex over emails / phone numbers / addresses / IP-shaped strings. Display name + user-id are scrubbed; authorHandle is set from the user’s preferred public handle.
S6	Per-user rate limit on publishing: max 50 published notes / week (matches the existing helpRateLimit shape). Editing counts as a new publish for limit purposes.
S7	Notes data is mineable: a nightly job exports (note, sessionConfig, outcome, eloDelta) tuples to ResearchLogExport (a normalized denormalized view in Postgres) that the reranker / fine-tune pipelines can read directly.

1.3 Out of scope for v1

Post-v1 enhancements (comments, voting, voice input, real-time collaboration, bulk markdown import, per-note hyperparameter hints) are tracked in `doc/Future_Ideas.md` → “Research Log — post-v1 enhancements”.

§2 Design

2.1 Schema

Two new Prisma models, additive migration. No backfill required — empty tables are fine on first deploy.

```
model TrainingSessionNote {
  id          String    @id @default(cuid())
  userId      String
  user        User      @relation(fields: [userId], references: [id])
  sessionId   String
  session     TrainingSession @relation(fields: [sessionId], references: [id], onDelete:
↪ Cascade)
  parentNoteId String?
  parent      TrainingSessionNote? @relation("NoteChain", fields: [parentNoteId], references:
↪ [id])
  children    TrainingSessionNote[] @relation("NoteChain")

  body        String    @db.Text           // user-authored markdown, ≤ 2 KB
  outcome     NoteOutcome
  tags        String[]
  // normalized lower-snake

  sharedWithCommunity Boolean @default(false)
  publishedAt  DateTime?
  helpDocId    String? @unique             // FK into the mirrored HelpDoc (when published)

  createdAt    DateTime @default(now())
  updatedAt    DateTime @updatedAt

  @@index([userId, createdAt])
  @@index([sessionId])
  @@index([sharedWithCommunity, publishedAt])
}

model ResearchLogEntry {
  id          String    @id @default(cuid())
  userId      String
  user        User      @relation(fields: [userId], references: [id])

  title       String           // ≤ 120 chars
  body        String    @db.Text // ≤ 4 KB
  tags        String[]
  category     ResearchEntryCategory // PLANNING | RETROSPECTIVE | OBSERVATION | OTHER

  sharedWithCommunity Boolean @default(false)
  publishedAt  DateTime?
  helpDocId    String? @unique

  createdAt    DateTime @default(now())
  updatedAt    DateTime @updatedAt

  @@index([userId, createdAt])
  @@index([sharedWithCommunity, publishedAt])
}

enum NoteOutcome {
  SUCCESS
  PLATEAU
}
```

```

REGRESSION
INCONCLUSIVE
}

```

```

enum ResearchEntryCategory {
    PLANNING
    RETROSPECTIVE
    OBSERVATION
    OTHER
}

```

Why two tables instead of one: - TrainingSessionNote is tightly coupled to a TrainingSession (FK + delete behavior + retrieval-time JOIN to hyperparameters). Optional sessionId would muddy that. - ResearchLogEntry is standalone — planning posts before a session, retrospectives spanning weeks, observations on bracket UX. Different shape. - Both share the publish-to-corpus pathway; the publish handler dispatches on type but writes to the same HelpDoc table.

Why helpDocId UNIQUE on the note table: - Mirrors the note’s published state in the Help corpus 1:1. Editing a published note updates the linked HelpDoc; unpublishing deletes the HelpDoc and clears the FK. - Lets the privacy scrubber + chunker rerun idempotently on re-publish.

2.2 API surface

All routes under /api/v1/research. Auth required for every route except the public published list (which is also reachable via the Help System retrieval — see §2.4).

Method + Path	Purpose	Notes
POST /re-search/sessions/:sessionId/notes	Create a new note on a training session	Body: { body, outcome, tags? }. Caller must own the session.
PATCH /research/notes/:noteId	Edit a note	Body: any of { body, outcome, tags }. Re-runs PII scrub if sharedWithCommunity.
DELETE /research/notes/:noteId	Delete a note	If published, also deletes the linked HelpDoc.
POST /research/notes/:noteId/publish	Toggle community-share on	Idempotent. Runs the PII scrubber, creates / updates the HelpDoc row, chunks + embeds via the existing Sprint-2 pipeline.
POST /re-search/notes/:noteId/unpublish	Toggle community-share off	Idempotent. Deletes the linked HelpDoc (+ its chunks).
GET /research/notes	List the caller’s notes	Filters: ?algo=rithm=...&outcome=...&tag=...&since=...&limit=... Paginated.
GET /research/notes/:noteId	Single note	403 unless owner or note is published.
POST /research/entries	Create an ad-hoc ResearchLogEntry	Body: { title, body, tags?, category }. Same publish-rescrub rules.
PATCH /research/entries/:entryId	Edit	Same publish-cascade rules.
DELETE /research/entries/:entryId	Delete	
POST /re-search/entries/:entryId/publish	Publish to community	—
POST /re-search/entries/:entryId/unpublish	Unpublish	—
GET /research/entries	List the caller’s entries	Filters: ?category=...&tag=...&since=...&limit=...
GET /research/community	Public list of published notes + entries	Read-only. Surfaces in the Profile journal under a “Community” tab. Rate-limited.

Method + Path	Purpose	Notes
GET /research/export.md	One-shot markdown export of all the caller's notes + entries	Server returns text/markdown with YAML front-matter per item.

2.3 PII scrubbing on publish

backend/src/services/research/pii.js — single pure function scrubForPublish(text):

- Regex strip of: email addresses, phone numbers (loose international match), IPv4 / IPv6, US-style street addresses (rough match — <num> <Word>+ (Street|Ave|Rd|Blvd|...)), credit-card-shaped digit runs.
- Heuristic detection of personal names is intentionally **not** attempted in v1 — too high a false-positive rate. The publish flow shows a preview with all matches highlighted so the user can edit before confirming.
- The user's own displayName and username are preserved if they choose attributeAs: 'handle' (the **default**); replaced with Anonymous if they explicitly toggle attributeAs: 'anonymous' in the publish modal.

UI: the publish modal shows a side-by-side preview (left: original, right: scrubbed). Confirm publishes; Cancel keeps the note private.

2.4 Help System integration

The Help System's retrieval pipeline (Sprint 2 §3 services/help/retrieval.js) currently returns top-k chunks from HelpChunk filtered by HelpDoc.publishedAt IS NOT NULL. Extension:

- **Add a source filter** to the retrieval query. Already supported by the schema (HelpDoc.source is a free-text label).
- **Three retrieval lanes** mixed at answer-build time:

Lane	HelpDoc.source value	Visibility	Default budget
corpus	'guide' (existing curated docs)	All users	4 chunks
private-notes	'private-note'	Only the asking user	2 chunks
community-notes	'community-note'	All users	2 chunks

The budget is the **upper bound** per lane — the existing relevance threshold still applies, so a low-quality private note gets cut even when there's room for it.

- **Citation format:** each chunk carries its HelpDoc.id + HelpDoc.source through the existing retrieval response. The answer renderer (HelpAnswer.jsx) gains a small badge: Guide, Your note, Community note by @handle.
- **Privacy filter at write-time, not query-time:** the private-note HelpDoc rows carry the author's userId in HelpDoc.ownerId (new optional column, additive migration). The retrieval query adds WHERE source = 'private-note' AND ownerId = \$callerId for the private lane. Community-note rows have ownerId IS NULL (anonymized) or ownerId = author (handle-attributed) — either way they're publicly readable.
- **shareNotesWithGuide preference:** stored as a key in the existing User.preferences Json @default("{}") column (no schema change). When false, the private-notes lane is skipped entirely for that user's queries. Defaults to true (absence-of-key == ON) because the user has already opted into the platform.

2.5 Data-mining export

Cron job backend/src/jobs/researchLogExport.js, runs nightly at 03:00 UTC. Builds a denormalized view-table ResearchLogExport:

```
model ResearchLogExport {
  id          String   @id @default(cuid())
  noteId      String?  @unique
  entryId     String?  @unique
  userId      String
  algorithm   String?  // copied from TrainingSession at export time
  hyperparameters Json? // snapshot at session end
```

```

preBenchmark    Float?
postBenchmark   Float?
eloBefore       Int?
eloAfter        Int?
eloDelta        Int?
noteText        String @db.Text
outcome         NoteOutcome?
category        ResearchEntryCategory?
tags            String[]
publishedAt     DateTime?
exportedAt      DateTime @default(now())
}

```

Why a denormalized table instead of a query view: training pipelines pull rows in batches; a static table avoids re-JOINing TrainingSession + BotSkill + GameElo for every batch. The export is rebuilt nightly (truncate + insert) so it’s eventually-consistent with the source tables — perfectly fine for offline training jobs.

The reranker (Sprint-5+ in Future_Ideas.md) reads from this table directly. Future fine-tune jobs ingest it via pgcopy. No new infrastructure needed beyond the cron.

2.6 UI surfaces

Sessions tab (GymPage.jsx → Sessions panel):

- Each session row gains an expandable “Notes” drawer (collapsed by default).
- Inside the drawer: existing notes **newest first by default**, with a small sort-toggle button in the drawer header that flips to oldest-first. Each row shows body / outcome pill / tag chips / publish badge / edit + delete affordances.
- “+ Add note” button at the bottom opens an inline editor: textarea, outcome dropdown, tags input (free-form, lower-snake normalized, typeahead from user’s own most-used + top community tags).
- Publish button on a saved note opens the side-by-side scrubber preview modal **only when SystemConfig.researchLog.publishEnabled === true** (Sprint 2 feature flag — see C6 in §7); otherwise the button is hidden entirely (Sprint 1 behavior).

Profile → Training journal tab (new):

- Section 1: **Session notes** — all TrainingSessionNote rows for the caller, filterable by algorithm / outcome / tag / date range. Click a row → expand inline editor or jump to the source session.
- Section 2: **Ad-hoc entries** — all ResearchLogEntry rows, filterable by category / tag / date.
- Section 3: **Community feed** — GET /research/community paginated. Filter by tag. Click a published note → read view. Cross-link to the same feed in the Guide drawer (see below).
- Right-rail: export to .md button (calls GET /research/export.md).
- Settings cog: shareNotesWithGuide toggle.

Guide drawer (GuidePanel.jsx):

- Two integration points:
 1. **Citation rendering in HelpAnswer.jsx** — the existing citation block gains a small icon prefix per source type and links each citation to its source (Guide doc, your own journal entry, or the published community note).
 2. **New “Community notes” section in the browse panel** — paginated list of published notes alongside the existing /help/:slug curated docs. Same GET /research/community API as the Profile sub-tab; shared component, two mount points. Cross-link header: “View in Profile journal →” / “View in Guide →”.

§3 Implementation plan

Three sprints, each independently shippable. Sprint 1 delivers user value standalone; Sprint 2 adds the Profile surface; Sprint 3 wires up Help retrieval.

Sprint 1 — Foundation (Sessions tab + CRUD)

Goal: users can write, edit, and delete notes attached to training sessions, visible inline in Gym → Sessions.

Backend:

1. **Schema** — TrainingSessionNote + NoteOutcome enum. Prisma migration. Tests: schema round-trip, FK + cascade behavior, unique constraint on helpDocId.
2. **Routes** — POST /research/sessions/:sessionId/notes, PATCH /research/notes/:noteId, DELETE /research/notes/:noteId, GET /research/notes, GET /research/notes/:noteId. All gated by requireAuth. Owner check uses the **denormalized note.userId** field (req.auth.userId === note.userId) — TrainingSession itself has no direct userId (ownership flows TrainingSession.modelId → BotSkill.createdBy), so on **create** we resolve ownership once via session.model.createdBy and stamp userId onto the note; all subsequent mutations check the stamped field, avoiding a per-write JOIN.
3. **Body validation** — body ≤ 2 KB, outcome must be a NoteOutcome value, tags array of ≤ 10 lower-snake strings each ≤ 32 chars. Reject with 400 + clear error code.
4. **Tests** — 10 vitest cases on the routes (happy paths + owner check + size limit + outcome validation + missing-session 404).

Frontend (landing):

5. **api.research.{create,update,delete,list,get}** in landing/src/lib/api.js.
6. **SessionNotesDrawer.jsx** — collapsible drawer with notes list + inline editor.
7. **Integration in GymPage.jsx** Sessions tab — render <SessionNotesDrawer sessionId={s.id} /> per row.
8. **Vitest** — 5 cases on the drawer (render, add, edit, delete, optimistic update + rollback on error).

Exit criteria: - A user can add a note to any of their training sessions and see it inline on next render. - A user cannot add a note to another user's session (returns 403 in the API; the button is hidden in the UI). - No publish flow yet — every note is private; the publish button on the row says "(Sprint 2)".

Effort: ~2 dev days.

Sprint 2 — Profile journal + Publish flow

Goal: users can browse / filter / export their notes from the Profile tab and opt-in to publish individual notes to the community.

Backend:

1. **Schema** — ResearchLogEntry + ResearchEntryCategory enum. HelpDoc.ownerId (optional FK to User). Migration is additive.
2. **Feature flag** — seed SystemConfig.researchLog.publishEnabled = false (default off). All publish-related routes + UI gate on this; the entire Sprint 2 surface ships behind the flag so the publish flow is *built* in Sprint 2 but *activated* only when the flag is flipped (no redeploy needed for activation).
3. **PII scrubber** — backend/src/services/research/pii.js with regex set listed in §2.3. Pure function + 15 unit-test cases.
4. **Publish routes** — POST /research/notes/:noteId/publish, POST /research/notes/:noteId/unpublish, POST /research/entries/:entryId/publish, POST /research/entries/:entryId/unpublish. Each creates / updates / deletes the linked HelpDoc row, runs the existing services/help/chunker.js + services/help/embedClient.js pipeline, sets helpDocId. Idempotent. Returns 503 when publishEnabled === false.
5. **Entry routes** — POST /research/entries, PATCH /research/entries/:entryId, DELETE /research/entries/:entryId, GET /research/entries.
6. **Community list** — GET /research/community — paginated query over HelpDoc.source IN ('community-note') JOIN the source TrainingSessionNote / ResearchLogEntry for the attribution handle.
7. **Export route** — GET /research/export.md returns text/markdown with YAML front-matter per item.
8. **Rate limit** — extend helpRateLimit.js with a research-publish bucket: 50 publishes / week per user.
9. **Tests** — ~25 vitest cases covering PII scrub, publish round-trip, unpublish cascade, community list pagination, export round-trip, rate-limit enforcement.

Frontend (landing):

9. **Profile → Training journal tab** — mount inside the existing flat landing/src/pages/ProfilePage.jsx (no pages/profile/ subdir exists); extract the tab body to landing/src/components/research/TrainingJournalTab.jsx. Filters, list, ad-hoc entry composer, community feed sub-tab.
10. **Publish modal** — side-by-side scrubber preview (PublishNoteModal.jsx). Shows highlighted matches + Cancel /

Confirm.

11. **Export button** — anchor with `download="training-journal.md"` pointing at `GET /research/export.md`.
12. **Settings toggle** — `shareNotesWithGuide` in the user prefs panel.
13. **Vitest** — ~12 cases on the new components.

Exit criteria: - The Profile tab is fully functional standalone — user can browse, filter, edit, delete, publish, and export their notes. - Published notes appear in `GET /research/community` for any user to browse (both Profile sub-tab and Guide drawer Community section). - The Guide drawer browse panel gets a new “Community notes” section sharing the same component as the Profile sub-tab; the Guide retrieval citations themselves are unchanged (that’s Sprint 3). - With `SystemConfig.researchLog.publishEnabled = false`, every publish-related surface (button, modal, community feeds, publish API routes) is hidden / 503s — Sprint 2 ships **dark**, awaiting an explicit flag flip.

Effort: ~3.5 dev days (bumped from 3 to cover the second community-feed mount + feature-flag plumbing).

Sprint 3 — Help retrieval integration

Goal: the Guide cites the user’s own notes (and community-published notes) when answering training questions.

Backend:

1. **Retrieval mixer** — refactor `services/help/retrieval.js` to take a `lanes: { corpus: 4, privateNotes: 2, communityNotes: 2 }` parameter and return chunks tagged with their lane. Default lanes wired from `SystemConfig` so we can tune without a deploy.
2. **Private-lane filter** — `WHERE source = 'private-note' AND ownerId = $callerId`. Community-lane filter: `WHERE source = 'community-note'`. Corpus lane: existing default.
3. **Citation metadata** — pass `source + ownerId + linked noteId / entryId` through to the `/help/ask` response so the renderer can show the right badge + deep-link.
4. **Preference gate** — read `user.shareNotesWithGuide`; if false, skip the private lane.
5. **Tests** — ~10 vitest cases: lane budget enforcement, private-lane owner filter, preference gate skip, citation metadata round-trip.

Frontend (landing):

6. **HelpAnswer.jsx citation block** — render the source-type badge + deep-link. Hover tooltip: “From your note on 2026-04-12” or “From a community note by @alice”.
7. **Vitest** — 3 cases on the renderer (private badge, community badge, attribution scrubbed when `ownerId IS NULL`).

Nightly export job:

8. **backend/src/jobs/researchLogExport.js** — runs at 03:00 UTC. Truncates + inserts into `ResearchLogExport`. Vitest covers the JOIN + truncate semantics. Wire into the existing `scheduledJobs.js` dispatcher.
9. **Schema** — `ResearchLogExport` model. Migration additive.

Exit criteria: - Asking the Guide a training question (e.g. “what’s a good learning rate for q-learning”) returns an answer that cites the user’s most-recent relevant note alongside the curated corpus docs. - A user who toggles `shareNotesWithGuide` off sees no private-note citations. - Community notes appear with the correct handle (or “Anonymous” when scrubbed). - The nightly export job populates `ResearchLogExport` with the expected JOIN shape.

Effort: ~3 dev days.

§4 Forward-compatibility — what Help Sprint 5+ inherits

This plan is designed to ship the Research Log **before** the Help System Sprint 5+ work in `Future_Ideas.md`. Sprint 5+ then inherits a corpus, a training-pair table, and a retrieval architecture pre-shaped for its requirements — no separate planning needed for the items listed below.

4.1 What each Sprint 5+ item gets for free

Sprint 5+ item	What Research Log delivers
Reranker (v2)	The ResearchLogExport denormalized table is pre-shaped for offline batch training — (prose, hyperparams, outcome, eloDelta, lane, feedback) tuples ready for pgcopy. The 3-lane retrieval mixer in services/help/retrieval.js is the natural seam — replace “budget per lane” with “rerank across lanes” once trained, no other refactor needed. Per-lane citation metadata in HelpFeedback means thumbs land on lane-tagged sources, giving the reranker lane-mix signal not just per-doc signal.
Embedding upgrade	Roughly doubles the corpus with diverse user prose (vs the curated Guide tone). The auto-reindex-on-model-change path from Help Sprint 2 already handles drop-in swaps; community-note chunks reindex via the same pipeline. Larger / more-diverse corpus = easier to measure the quality delta of a new embedding model.
Gap-filling from real traffic	Published community notes are gap-filling in a different shape — bottom-up rather than admin-curated. The /admin/help/metrics dashboard can grow a “most-cited community notes” panel using existing infrastructure. The curation queue at /admin/help/queries handles moderation of community-flagged notes via the same UI.
LoRA fine-tune (v3)	Single biggest unlock. The ResearchLogExport rows are literally the training-pair shape: (prose context, structured params, outcome label). The fine-tune can learn what makes an answer HELPFUL in <i>training contexts</i> specifically, not just syntactically good. Pre-Research-Log there’s no such labelled corpus; post-Research-Log it accrues automatically. The Sprint-5+ “~5k HELPFUL pairs” threshold becomes a function of Research Log adoption rate, not a separate data-collection effort.
Streaming filter / Voice input / Cross-session help history / Per-IP rate limit / PII auto-redact	Orthogonal — no dependency, no benefit. Ship whenever.

4.2 Architectural seams deliberately left open

Each of these is a choice made *in the Research Log plan* specifically so Sprint 5+ slots in cleanly:

1. **3-lane retrieval mixer** (services/help/retrieval.js, Sprint 3 §1): the reranker replaces “budget per lane” with “rerank across lanes” by changing only that one function. The API contract above (the /help/ask handler) and below (the chunk store) doesn’t change.
2. **HelpDoc.source discriminator**: already free-text. Adding 'private-note' and 'community-note' (Sprint 2) doesn’t require schema changes for the reranker; it just adds more rows the reranker scores.
3. **HelpDoc.ownerId** (new in Sprint 2): the optional FK enables per-user retrieval beyond Research Log too — Sprint 5+ items like personalized canonical-doc ranking or user-specific corpus subsets get this for free.
4. **ResearchLogExport nightly job** (Sprint 3 §8): built for offline consumers. Reranker / fine-tune jobs read from it directly, no new ETL needed. The truncate-and-insert pattern means schema evolution is safe — add columns without backfill, the next nightly run rebuilds.
5. **Per-lane citation metadata in the SSE stream** (Sprint 3 §3): once the answer renderer shows lane badges, feedback on the cited chunks is implicitly lane-tagged. The thumbs-down data the reranker eventually trains on already carries the lane signal.

4.3 What Sprint 5+ still needs fresh planning for

The Research Log doesn’t eliminate Sprint 5+ planning — it changes what’s left to decide:

- **Embedding-model selection criteria:** corpus characteristics (size, prose diversity, technical-vocabulary density) matter for picking BGE-small vs a larger-dim model. Re-evaluate once the community-note corpus has ~3–6 months of accumulation.
- **LoRA fine-tune compute budget + vendor:** still an open decision (own GPU vs hosted via Together / Anyscale / OpenAI fine-tune API). The data shape is settled; the training-cost / latency / vendor-lock trade-off is not.
- **Streaming filter implementation:** token-level vs chunk-level. Orthogonal to Research Log; same planning effort either way.

§5 Sprint checklist

Sprint 1 — Foundation (~2 days) — ☒ SHIPPED on dev 2026-05-16 (commit 11baa14)

- ☒ **Schema:** add TrainingSessionNote + NoteOutcome to packages/db/prisma/schema.prisma
- ☒ Generate + apply migration locally (docker compose run --rm backend npx prisma migrate deploy — migrate dev wanted a reset because earlier migrations had been hand-edited, so we authored 20260516120000_research_log_notes/migration.sql directly and applied via deploy)
- ☒ **Route file:** backend/src/routes/research.js mounted at /research in index.js
- ☒ Implement + test POST /research/sessions/:sessionId/notes
- ☒ Implement + test PATCH /research/notes/:noteId
- ☒ Implement + test DELETE /research/notes/:noteId
- ☒ Implement + test GET /research/notes (with outcome / tag / since / limit filters — algorithm filter deferred to Sprint 2 where it can JOIN through to TrainingSession.model.algorithm)
- ☒ Implement + test GET /research/notes/:noteId
- ☒ Tests: owner check (403 on cross-user mutation), body size limit (≤ 2 KB), invalid outcome (400), missing session (404). **Shipped 13 vitest cases** (target was 10).
- ☒ **Client API:** api.research.{createNote, updateNote, deleteNote, listNotes, getNote} in landing/src/lib/api.js
- ☒ **Component:** landing/src/components/research/SessionNotesDrawer.jsx
- ☒ Integrate into landing/src/pages/GymPage.jsx Sessions tab (mounted from landing/src/components/gym/SessionsTab.jsx under the selected-session detail panel)
- ☒ Vitest: 5 cases on the drawer (render, add, edit, delete, optimistic + rollback)
- ☒ Manual QA: write a note → reload → still there; edit → reflects; delete → gone (2026-05-16, project lead)
- ☒ Update doc/V1_Acceptance.md with the new flow (Stage 12)

Sprint 2 — Profile journal + Publish flow (~3.5 days) — shipped 2026-05-16

- ☒ **Schema:** add ResearchLogEntry + ResearchEntryCategory + HelpDoc.ownerId + HelpDoc.source (defaulting to 'guide' for all existing rows)
- ☒ Migration: 20260516200000_research_log_entries_and_publish
- ☒ **Feature flag:** SystemConfig.researchLog.publishEnabled (absence-of-row === OFF, no seed). Gate every publish route on the flag (return 503 when off). Server-side gate is the source of truth; the frontend modal surfaces a friendly 503 message.
- ☒ **PII scrubber:** backend/src/services/research/pii.js + 16 vitest cases
- ☒ **Publish routes** (notes + entries) — wire into existing services/help/chunker.js + services/help/embedClient.js. Title scrubbed too so HelpDoc.title can't leak.
- ☒ **Unpublish cascade** — delete linked HelpDoc + chunks; clear helpDocId
- ☒ **Entry CRUD routes** (POST/PATCH/DELETE/GET /research/entries[...]) with title ≤ 120 , body ≤ 4 KB, category enum
- ☒ **Community list:** GET /research/community — cursor pagination on (createdAt, id), tag filter, attribution via HelpDoc.owner JOIN (null-safe when user deleted)
- ☒ **Export:** GET /research/export.md returns markdown with YAML front-matter per item, interleaved reverse-chrono
- ☒ **Rate limit:** new middleware/researchPublishRateLimit.js — 50 publishes / week / user. Unpublish is not gated. CLI-bypass honored.
- ☒ Tests: 96 backend cases (publish 17, scrub 16, route 62, community service 11, rate-limit 6, config 5) — exceeds the original 25-case target. Feature-flag gate proven in both service and route tests.
- ☒ **Page:** landing/src/components/research/TrainingJournalTab.jsx mounted inside flat landing/src/pages/ProfilePage.jsx as a new accordion section. Three sub-tabs: My notes / My entries / Community.

- **Guide drawer integration:** community section inside the Guide browse panel — deferred to Sprint 3 (lands naturally with the retrieval lane work)
- ☒ **Modal:** `landing/src/components/research/PublishNoteModal.jsx` (side-by-side original/scrubbed diff, redaction chips per kind, confirm-checkbox gate). Client-side preview mirror lives at `landing/src/lib/research/pii.js`.
- ☒ **Settings toggle:** `shareNotesWithGuide` stored as a key inside the existing `User.preferences` `Json` column (no schema column); surfaced as a “Share with Guide” toggle in the Training Journal header. Backed by `GET/PATCH /research/preferences`.
- ☒ Landing tests: 21 cases on the new components (pii 4, modal 7, tab 10) — exceeds the 14-case target.
- ☒ Manual QA with flag OFF (absence-of-row): publish endpoints 503; the modal surfaces “Community publishing is currently disabled.”
- ☒ Manual QA with flag ON: write → publish → community feed renders → unpublish → gone from feed but still in own journal.
- ☒ Update `doc/Guide_Operations.md` §5.9 with the `publishEnabled` flip procedure + community-lane purge runbook

Sprint 3 — Help retrieval integration (~3 days) — shipped 2026-05-16

- ☒ **Retrieval refactor:** `searchLanes()` in `backend/src/services/help/helpService.js` accepts a `lanes` budget map + `userId` + `shareWithGuide` and returns lane-tagged chunks. (No separate `retrieval.js` extraction — the lane logic lives next to the existing FTS/vector branches it composes; a follow-up split is cheap if it grows.)
- ☒ **Default lanes from SystemConfig** — `help.lanes.corpus/help.lanes.privateNotes/help.lanes.communityNotes` (defaults 4 / 2 / 2). `getLaneBudgets()` reads the row and clamps each lane to `[0, 25]`.
- ☒ **Private-lane owner filter** at the SQL level (not app code) + preference gate via `getUserSharePref()` reading `User.preferences.shareNotesWithGuide`. Lane skipped when `userId` null OR pref off OR pref read errors (fail-safe).
- ☒ **Citation metadata** plumbed through `/help/ask SSE` — done frame now carries citations: `{ docId, lane, slug, title, author }`; `buildCitations()` dedupes by `docId` with priority `privateNotes > communityNotes > corpus`.
- ☒ Tests: 20 vitest cases (target 10) on lanes, budgets, owner filter, preference gate, citation metadata round-trip, dedupe.
- ☒ **Renderer:** badge + deep-link in `landing/src/components/guide/HelpAnswer.jsx` — one citation chip per source with `Guide / Your note / Community` icons and `data-lane` for styling/analytics.
- ☒ Vitest: 3 cases on the citation rendering (rendering, empty-array hides the block, `data-lane` attribute per lane).
- ☒ **Schema:** `ResearchLogExport` (materialised view, truncate-and-insert) + migration `20260517000000_research_log_exports` + nightly job `backend/src/jobs/researchLogExport.js`.
- ☒ Wire into startup via `startResearchLogExportCron()` — hourly `setInterval` gated on UTC hour 3 + per-UTC-day idempotency. (Used the existing `setInterval` cron pattern from `metricsSnapshotService.js` rather than the `scheduled-Jobs` queue — same effect, less wiring; the queue route is a cheap migration later if admin visibility into the next run is wanted.)
- ☒ Tests: 7 vitest cases on `TRUNCATE+INSERT` atomicity, discriminator stamping, empty-table no-op, cron tick gating.
- ☒ Manual QA scaffold: `runResearchLogExport()` is callable from the Node REPL inside the container if the user wants to ad-hoc trigger; cron auto-runs on the next 03:00 UTC after boot.
- ☒ Added `doc/Help_Corpus/community-notes.md` — one-pager explainer of what gets shared, what stays private, how the lanes + citations work, and the rate limit. Auto-seeded into the live corpus on next backend boot via `seedCorpus()`.

Deferred to a focused follow-up (see `doc/Future_Ideas.md` → “Private-notes auto-indexing pipeline”):

- Mirroring unpublished notes/entries into `source='private-note'` `HelpDocs` so the `privateNotes` lane actually returns content. Sprint 3 wired the lane end-to-end (filter, citations, UI) but the lane is empty until that pipeline ships. Lift is ~2 days; touches Sprint 1+2 code paths (note create/update hooks + a backfill job).

§6 Risks + mitigations

Risk	Likelihood	Impact	Mitigation
PII slips through the scrubber	Medium	High (privacy)	Side-by-side preview before publish; user must actively confirm. Server-side scrub on every edit. Quarterly review of the regex set against published-note samples.
Community-note corpus gets noisy / spam	Medium	Medium	Per-user rate limit (50 / week). HELP_ADMIN moderation queue (reuse the existing /admin/help/queries curation infrastructure) for community notes flagged by feedback thumbs-down.
Private-lane retrieval leaks across users	Low	Critical (privacy)	Owner filter applied at the SQL level, not in app code. Test asserts cross-user query returns zero private chunks. Annual penetration-style review.
Retrieval lane budget drowns curated content	Low	Medium	Curated lane wins ties (default 4 vs 2/2). Reranker (Sprint-5+) re-scores across lanes once trained.
Nightly export grows unbounded	Low	Low	ResearchLogExport is a truncate-and-insert view — its size is bounded by source tables. Notes themselves have row-level TTL governance via the existing pruneIfNeeded job.

§7 Decisions locked (2026-05-16 walkthrough)

All v1 decisions confirmed in a planning walkthrough with the project lead. Snapshot below for future readers.

Privacy & security

#	Decision	Locked
A1	shareNotesWithGuide preference default	ON — private notes mix into Help retrieval by default; toggle in Profile settings to disable
A2	Community-note moderation model	post-hoc takedown — notes go live immediately; /admin/help/queries curation queue serves as reactive review; 50/week rate limit + thumbs-down feedback flag spam
A3	PII scrubber — personal-name detection	skip in v1 — emails / phones / IPs / addresses / cards stripped; names left to user via side-by-side preview before confirm

#	Decision	Locked
A4	Attribution default on publish	handle — user’s public handle by default; anonymous is opt-in via the publish modal dropdown

UX

#	Decision	Locked
B1	Schema split for notes vs entries	two tables — TrainingSessionNote (FK to TrainingSession) + ResearchLogEntry (standalone)
B2	Community-feed location	both surfaces — Profile → Training Journal → Community sub-tab AND a new “Community notes” section in the Guide drawer browse panel; shared GET /research/community API
B3	Note ordering on a session row	newest first by default , with a per-tab sort toggle in the drawer header
B4	Post-v1 enhancements (comments, voting, voice, real-time collab, bulk import, hyperparameter hints)	deferred — tracked in doc/Future_Ideas.md → “Research Log — post-v1 enhancements”

Implementation details

#	Decision	Locked
C1	Markdown rendering library	react-markdown — reused from Help System; supports code blocks + GFM tables
C2	Tag taxonomy	free-form — normalized to lower-snake-case + dedup; typeahead of user’s own most-used + top community tags
C3	Published-then-deleted note behavior	hard cascade — HelpDoc + chunks deleted; historical citations render as “(note removed by author)”
C4	Body size limits	2 KB notes / 4 KB entries
C5	Publish rate limit	50 / week per user (no daily cap); reuses the helpRateLimit.js shape
C6	Sprint sequencing	Sprint 1 ships standalone with publishing hidden. Sprint 2 builds the publish flow behind SystemConfig.researchLog.publishEnabled — when the flag flips, the publish button appears; flag flip is the activation, not a redeploy
C7	Help retrieval lane budgets	4 corpus / 2 private-notes / 2 community-notes, curated wins ties; all readable from SystemConfig (help.lanes.corpus, help.lanes.privateNotes, help.lanes.communityNotes) so they tune without a deploy

Sequencing

#	Decision	Locked
D1	Research Log vs Help System Sprint 5+ ordering	Research Log first. Sprint 5+ items (reranker, embedding upgrade, gap-filling, LoRA fine-tune) all inherit corpus + training-pair data from the Research Log work — see §4 for inheritance details

§8 Cross-references

- doc/Future_Ideas.md — Research Log entry (this plan's source)
- archive/Help_System_Plan.md — retrieval pipeline, chunker, embedder (Sprint 2 §3)
- archive/Help_System_Sprint_Tracker.md — corpus + admin curation patterns to mirror
- doc/V1_Acceptance.md — QA flows to extend with Research Log paths once Sprints 1 + 2 ship
- doc/Observability_Plan.md — metrics to add (researchNotesCreated, researchNotesPublished, researchCitationsServed)